Natacha Spitzer

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Citizenship: Canada, France, and United States

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Education

University of Pennsylvania 2021 – Present
Ph.D. in Economics, expected May 2027 (Advisor: George Mailath)

Paris School of Economics 2019 – 2021
Master of Analysis and Policy in Economics (Summa Cum Laude)

University of Rennes I 2016 – 2019
Bachelor of Economics and Management (Summa Cum Laude)

Exchange Program at the University of Rochester 2018 – 2019

Research Interests

Primary Field: Microeconomic Theory
Secondary Fields: Industrial Organization, Organizational Economics
Interests: Contract Theory, Information Economics, Experimentation and Learning, Repeated Games, Networks, Mechanism Design

Research

Contracting with Unknown Complementarity: Ratchets and Replacement Risk

Job Market Paper

Abstract: How do a firm and a worker contract while learning how productive their relationship is? Effort simultaneously generates value for the firm and reveals information about the fit between the worker and his assignment. But the prospect of future wage adjustments and replacement distorts his incentives to experiment. I study these forces in a two-period principal–agent model with moral hazard, limited commitment, and limited liability. I first establish existence of a pure-strategy equilibrium and characterize the optimal spot contract: when there is a chance that effort is counterproductive, the agent may be rewarded following both sufficiently strong and sufficiently poor performance, as both are evidence of effort. After the first period, the principal decides whether to replace the agent and, conditional on retention, offers a new wage contract. A forward-looking agent therefore has an incentive to influence her belief, and because he captures limited-liability rents in each period, replacement is costly to him. This incentive decomposes into an *extensive* retention force and an *intensive* wage force. Relative to a benchmark without belief manipulation, the retention force dominates when the parties are initially pessimistic about the fit, leading to over-experimentation, while the ratchet dominates when they are initially optimistic and effort is not too costly, leading to under-experimentation. Dynamic incentives thus raise experimentation and learning when the agent's job is at risk and lower it once it is secure.

Optimal Monitoring of AI Use with Reputation Concerns

Working Paper, with Mengjia Xia, under review

Abstract: We study how an accuracy-seeking manager should monitor workers' use of AI when workers care about their reputation. A worker of privately known ability chooses between costly effort and a tool. The manager may monitor his choice at a cost. We characterize equilibrium technology choices and then endogenize monitoring. The optimal policy is simple: either the manager does not monitor and everyone uses the tool, or she monitors just enough to sustain separation, in which high-ability workers exert effort and low-ability workers use the tool. Mandatory disclosure is never optimal, and optimal monitoring is

non-monotonic in the quality of the tool. Relative to the efficient benchmark, it can induce both over- and under-adoption. The policy has the same shape whether effort and the tool are substitutes or complements. When monitoring can depend on the outcome, the optimal audit targets success when able workers are rare and failure when they are common.

On the Irrelevance of Networks in Repeated Games
Working Paper, under review

Abstract: When do interaction and information networks affect equilibrium outcomes in environments with many bilateral repeated games? Using a multimarket Cournot counterexample with symmetric firms and markets, I show that the ability of all pairs of opponents to sustain collusion in equilibrium depends on both which players interact and what each player observes about others' interactions. I then develop a general framework and identify a sufficient condition on stage-game payoffs, *Optimality of Nash Reversion*, under which network structure is irrelevant: an action profile is sustainable between all opponents if and only if it is sustainable in the corresponding two-player game.

Selection in Networks with Imperfect Observability
Master's Thesis, Paris School of Economics (Advisor: Francis Bloch; Reader: Olivier Compte)

Abstract: I study ranking and selection in a network in which agents observe cardinal information about their neighbors that a social planner does not. Under perfect observability, the planner can construct an ex-post incentive compatible and efficient ranking if and only if each agent has at least two neighbors, and the sparsest such networks are graphs with cyclic components. Under imperfect observability, efficiency is lost in Bayesian equilibrium in triangles, but an approximately efficient mechanism exists when signals are precise; welfare under the optimal mechanism increases in signal precision and is higher in directed than in undirected triangles.

Presentations

Econometric Society World Congress (Seoul, 2025); NSF Network Science in Economics Conference (Stanford, 2025, poster); Pennsylvania Economic Theory Conference (Penn State, 2024 and 2025, posters); Advanced School in Economic Theory (Israel Institute for Advanced Studies, 2023, poster); Penn Micro Theory Lunch Seminar (2022–2026)

Organizing

Graduate Women in Economic Theory Conference, Yale University (co-organizer)	2026
Penn Micro Theory Lunch Seminar (organizer)	2024 – 2025
Microeconomic theory student reading group, University of Pennsylvania (organizer)	2023 – 2024

Other Research Experience

Research Assistant, Paris School of Economics 2019 – 2021
to Francis Bloch, Nicolas Jacquemet, Hillel Rapoport, and Akiko Suwa-Eisenmann

Research Intern, CNRS-CREM, University of Rennes I Summer 2019
Supervisor: Marie-Hélène Hubert

Teaching Experience

Instructor 2023 – Present
University of Pennsylvania

Game Theory	Summer 2025
Intermediate Microeconomics	Summer 2023

Teaching Assistant 2022 – Present
University of Pennsylvania

Introduction to Microeconomics, Dr. Anne Duchene	Fall 2026
Microeconomics for Managers, Prof. Kent Smetters	Summer 2026
Behavioral Economics, Prof. Kevin He	Fall 2025
Microeconomics for Managers, Prof. Kent Smetters	Summer 2025
Intermediate Microeconomics, Prof. George Mailath	Spring 2025
Industrial Organization, Prof. Aislinn Bohren	Fall 2024
Contract Theory, Prof. Steven Matthews	Spring 2024
Introduction to Microeconomics, Dr. Anne Duchene	Fall 2023
Mathematical Economics, Prof. Andrew Postlewaite	Spring 2023
Introduction to Microeconomics, Dr. Anne Duchene	Fall 2022

Head Tutor of Microeconomics 2017 – 2018
University of Rennes I

Awards

Sidney Weintraub Memorial Prize in Economics, University of Pennsylvania	2025
Robert and Anita Summers Graduate Research Award in Economics, University of Pennsylvania	2025
Fontaine Fellowship, University of Pennsylvania	2021

Additional Skills

Programming Languages: Python, R, Stata, MATLAB, Mathematica, LaTeX
Languages: English (Native), French (Native)

References

George Mailath (Advisor), Department of Economics, University of Pennsylvania
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Kevin He, Department of Economics, University of Pennsylvania
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Rakesh Vohra, Department of Economics, University of Pennsylvania
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